43. Cloud SDK Patterns: Config, Retries, Pagination

43. Cloud SDK Patterns: Config, Retries, Pagination

Phase: Phase 6 — Systems and infrastructure
Estimated time: 1–2 days
Slug: `43-cloud-sdk-patterns`

Learning objectives

- Configure clients via env/role credentials
- Retry with exponential backoff + jitter
- Paginate list APIs correctly
- Treat cloud APIs as unreliable networks

Prerequisites

Complete prior numbered lessons. Type every example yourself.

Retries

Retry transient errors (timeouts, 429, 503), not validation errors. Exponential backoff with jitter prevents synchronized stampedes.

```
func Retry(ctx context.Context, fn func() error) error {
    // backoff loop checking ctx.Done()
}
```

Pagination

Most list APIs return a cursor/token. Loop until empty. Bound max pages.

Config

Prefer IAM roles / workload identity over long-lived keys in files. Separate config loading from business logic for tests.

Common mistakes (especially from Python instincts)

- Infinite retries without deadline
- Retrying non-idempotent POSTs blindly
- Loading all pages into memory without need

Exercises

Work in `exercises/43-cloud-sdk-patterns/`. Read that folder's `README.md`.

- **A (Drill):** make unit tests pass with the simplest correct code.
- **B (Apply):** extend with an edge case called out in the exercise README.
- **C (Challenge):** rewrite your solution from memory the next day without looking.

Do not open `solutions/43-cloud-sdk-patterns/` until you have an honest failing attempt.

Self-check

1. What is jitter?
2. Which errors retry?
3. Why cursor pagination?

Further reading

- https://aws.github.io/aws-sdk-go-v2/docs/